Appendix 1 - Options Appraisal Matrix

R001	Asymchem PDF Extension Project																								
Option	Description	Existing Hydrogenation Building to remain	New PDF Building Extension to maximise available area	Central circulation corridor to allign	North & South scientific areas above and below central corridor are symetrical	Milestone Achievability (score 1 to 10) weighting 25%. Score 1-5 is highly complex, lots of interfaces, enabling works etc. Score 6-10 is low complexity, straightforward.			Impact on site operations (score 1 to 10) weighting 20% Score 1-5: Major impact requiring building shutdown and impact to science Score 6-10: Low impact and manageable planned shutdowns / no shutdowns			Project Complexity (score 1 to 10) weighting 20%. Score 1-5 is highly complex, lots of interfaces, enabling works etc. Score 6-10 is low complexity, straightforward.			Construction safety (score 1 to 10) weighting 15% 1-5 Low score - High H&S risk 6-10 High score - Low H&S risk			Requirements Compliance (score 1 to 10) weighting 20% Score 1-5: Major impact - User requirements Not met Score 6-10: Low impact - Uers requirement met			Total Score (Out of 10) Higher Score = Better Solution	URS Compliant	Pros	Cons	Cost
						Score	Weight	Total	Score	Weight	Total	Score	Weight	Total	Score	Weight	Total	Score	Weight	Total					
Existing Hydrogenation Building - Removed/Relocated																									
1	1. Existing Hydrogenation Building to be removed / relocated 2. New feasibility study required if Hydrogenation Building operations are still required - Where can the Hydrogenation be relocated to? 3. New PDF Extension building meets client requirements - Equipment & central corridor allignment	No	Yes	Yes	Yes	8	25%	2.00	8	20%	1.60	8	20%	1.60	9	15%	1.35	8	20%	1.60	8.15	Yes	1. Removal of Hydrogenation Building will improve constructability on new extension building 2. New PDF extension meets client's requirements	1. Existing Hydrogenation to be removed or relocated 2. Cost for Hydrogenation removal / relocation to be considered	£££
Existing Hydrogenation Building to Remain - New Extension Building to be Offset - Central Corridor Allignment																									
2A	1. Existing Hydrogenation builing to remain 2. New PDF Extension building to be offset from existing Hydrogenation building - Constructability & safety 3. New & existing central circulation corridors to allighn 4. New extension building will not be symetrical - North of the central corridor will be smaller than the south area	Yes	Yes	Yes	No	8	25%	2.00	10	20%	2.00	5	20%	1.00	5	15%	0.75	4	20%	0.80	6.55	No	1. Existing Hydrogenation to remain - No cost required to remove building	1. Hydrogenation building will increase the cost of the façade of the new PDF extension 2. Keeping the Hydrogenation building, moves the new PDF extension - to enable construction & for safety	£££
Existing Hydrogenation Building to Remain - New Extension Building to be Offset - Central Corridor Offset																									
2B	1. Existing Hydrogenation builing to remain 2. New PDF Extension building to be offset from existing Hydrogenation building - Constructability & safety 3. New & existing central circulation corridors do NOT allign 4. New extension building will be symetrical around central corridor	Yes	Yes	No	Yes	8	25%	2.00	10	20%	2.00	5	20%	1.00	5	15%	0.75	5	20%	1.00	6.75	No	1. Existing Hydrogenation to remain - No cosdt required to remove building	1. Hydrogenation building will increase the cost of the façade of the new PDF extension 2. Keeping the Hydrogenation building, moves the new PDF extension - to enable construction & for safety	£££
Existing Hydrogenation Building to Remain - New Extension Building to be Reduced in Size																									

3	1. Existing Hydrogenation builing to remain 2. New PDF Extension building to be offset from existing Hydrogenation building - Constructability & safety 3. New & existing central circulation corridors do allign 4. New extension building will be smaller than required	Yes	No	Yes	No	**8**	25%	2.00	**10**	20%	2.00	**5**	20%	1.00	**5**	15%	0.75	**3**	20%	0.60	**6.35**	**No**	1. Existing Hydrogenation to remain - No cost required to remove building	1.Keeping the Hydrogenation building will increase the cost of the façade of the new PDF extension - Protection agaist blasts/explosions from the Hydrogenation building 2. Hydrogenation building reduces the size of the new PDF extension	**££**
4	The first floor slab to be raised by 2000mm to enable the installation & operation of the 3No Hastelloy units on the ground floor	NA	NA	Yes	Yes	**8**	25%	2.00	**1**	20%	0.20	**2**	20%	0.40	**8**	15%	1.20	**1**	20%	0.20	**4.00**	**No**	Hastelloy units can be operated	1. Floor slabs between the extsing building and the new extension do not align 2. Additional stairs and lifts are required 3. New goods lift need to be reloacted to the far end of the new extension	**£££**